Task 5: Exploratory Data Analysis (EDA)

Data Analyst Internship – Elevate Labs

Dataset: Titanic Dataset

Objective: To extract meaningful insights using statistical and visual exploration with Python, Pandas, Matplotlib, and Seaborn.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

print("Libraries imported successfully!")
```

```
Libraries imported successfully!
```

```
df = pd.read_csv("Titanic.csv")

print("Dataset loaded successfully!")
df.head()
```

```
Dataset loaded successfully!
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S

Next steps: Generate code with df New interactive sheet

1. Understanding the Dataset

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   PassengerId  891 non-null    int64  
 1   Survived     891 non-null    int64  
 2   Pclass       891 non-null    int64  
 3   Name         891 non-null    object 
 4   Sex          891 non-null    object 
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64  
 7   Parch        891 non-null    int64  
 8   Ticket       891 non-null    object 
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object 
 11  Embarked     889 non-null    object 
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

```
df.describe()
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

```
df['Survived'].value_counts()
```

	count
Survived	
0	549
1	342

dtype: int64

```
df['Sex'].value_counts()
```

Sex	count
male	577
female	314

dtype: int64

```
df.isnull().sum()
```

	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

dtype: int64

2. Data Cleaning and Missing Value Handling

```
df['Age'] = df['Age'].fillna(df['Age'].median())
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])
df = df.drop('Cabin', axis=1)

print("Missing values handled successfully!")
```

```
Missing values handled successfully!
```

```
df.isnull().sum()
```

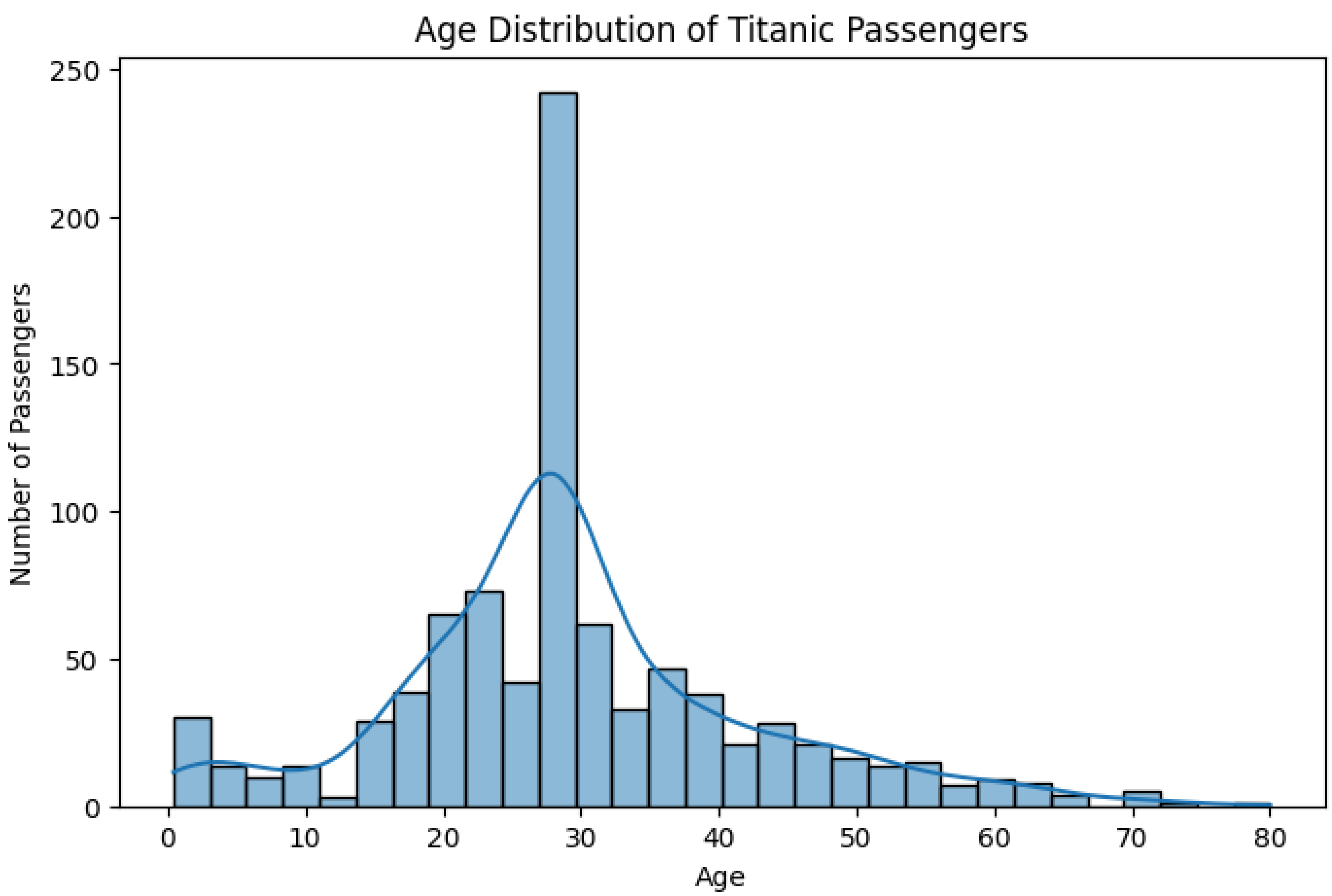
	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	0
SibSp	0
Parch	0
Ticket	0
Fare	0
Embarked	0

dtype: int64

3. Univariate Analysis

3.1 Age Distribution

```
plt.figure(figsize=(8, 5))
sns.histplot(df['Age'], bins=30, kde=True)

plt.title('Age Distribution of Titanic Passengers')
plt.xlabel('Age')
plt.ylabel('Number of Passengers')

plt.show()
```

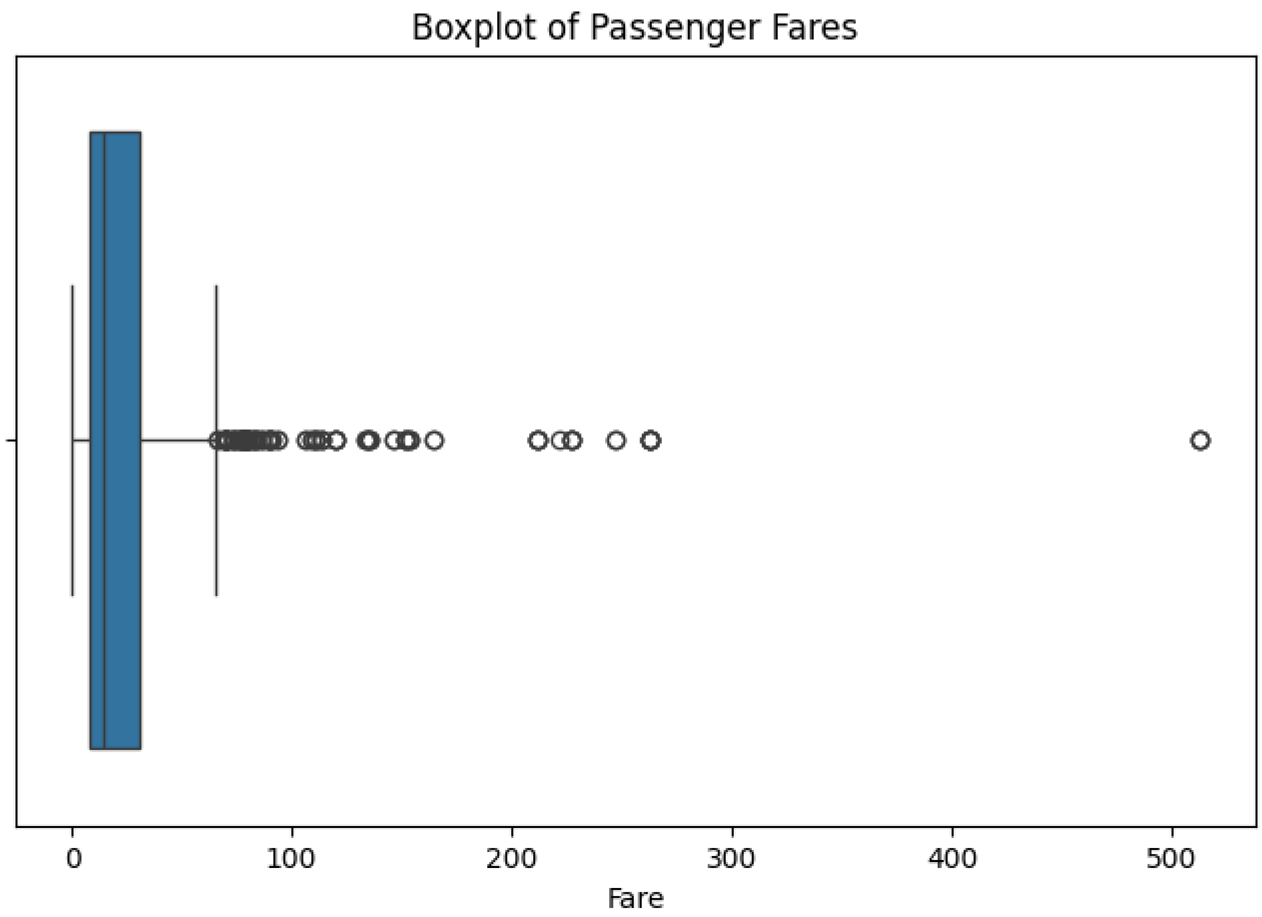


Observation: Most passengers were between 20 and 40 years old. The age distribution is slightly right-skewed, with fewer elderly passengers. The prominent peak around age 28 is partly due to missing age values being filled with the median age.

3.2 Fare Distribution and Outliers

```
plt.figure(figsize=(8, 5))
sns.boxplot(x=df['Fare'])
```

```
plt.title('Boxplot of Passenger Fares')
plt.xlabel('Fare')

plt.show()
```

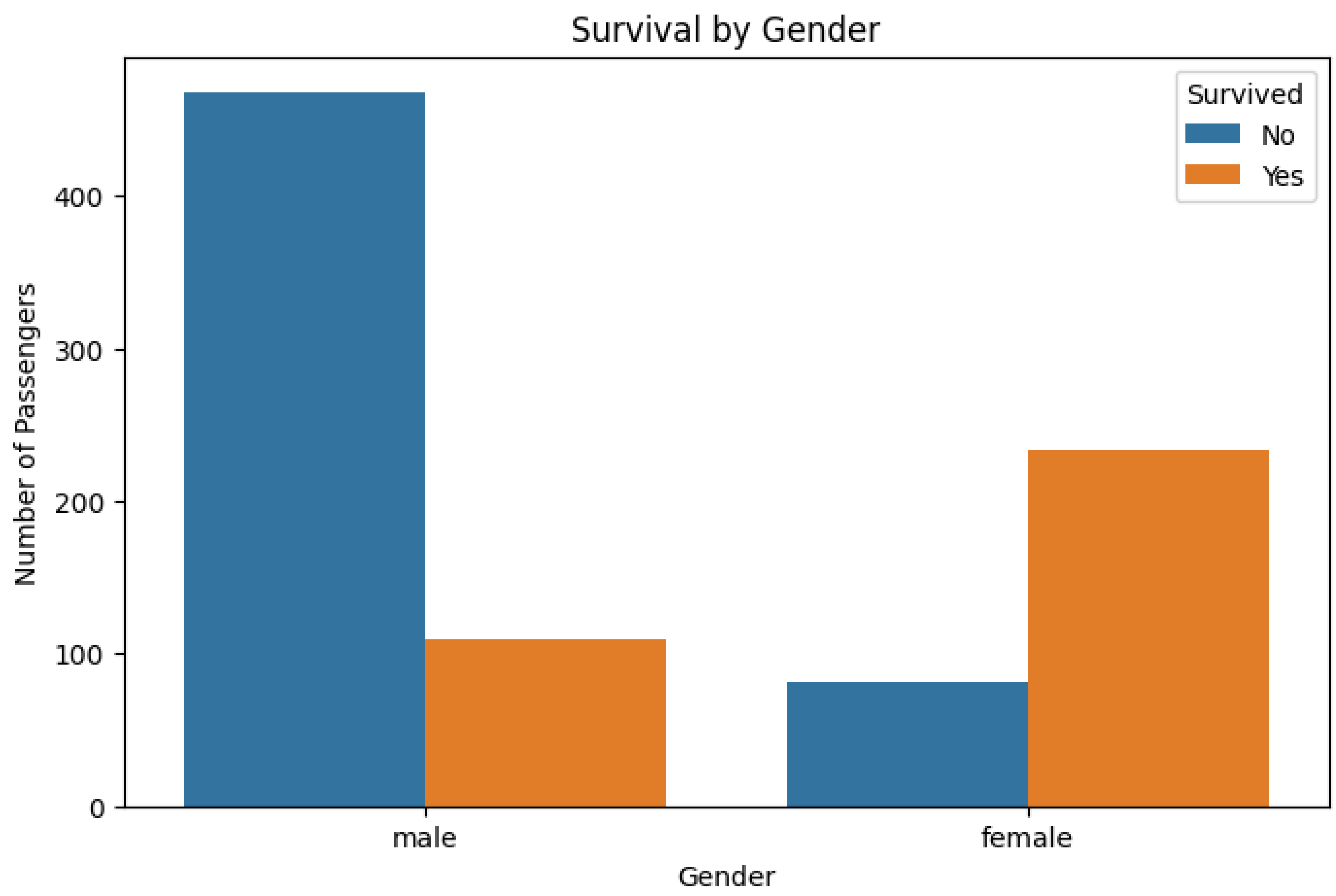


Observation: Most passengers paid relatively low fares, while several high-fare outliers are present. The fare distribution is strongly right-skewed, with the highest fare reaching about 512.

4. Bivariate Analysis

4.1 Survival by Gender

```
plt.figure(figsize=(8, 5))
sns.countplot(data=df, x='Sex', hue='Survived')

plt.title('Survival by Gender')
plt.xlabel('Gender')
plt.ylabel('Number of Passengers')
plt.legend(title='Survived', labels=['No', 'Yes'])

plt.show()
```

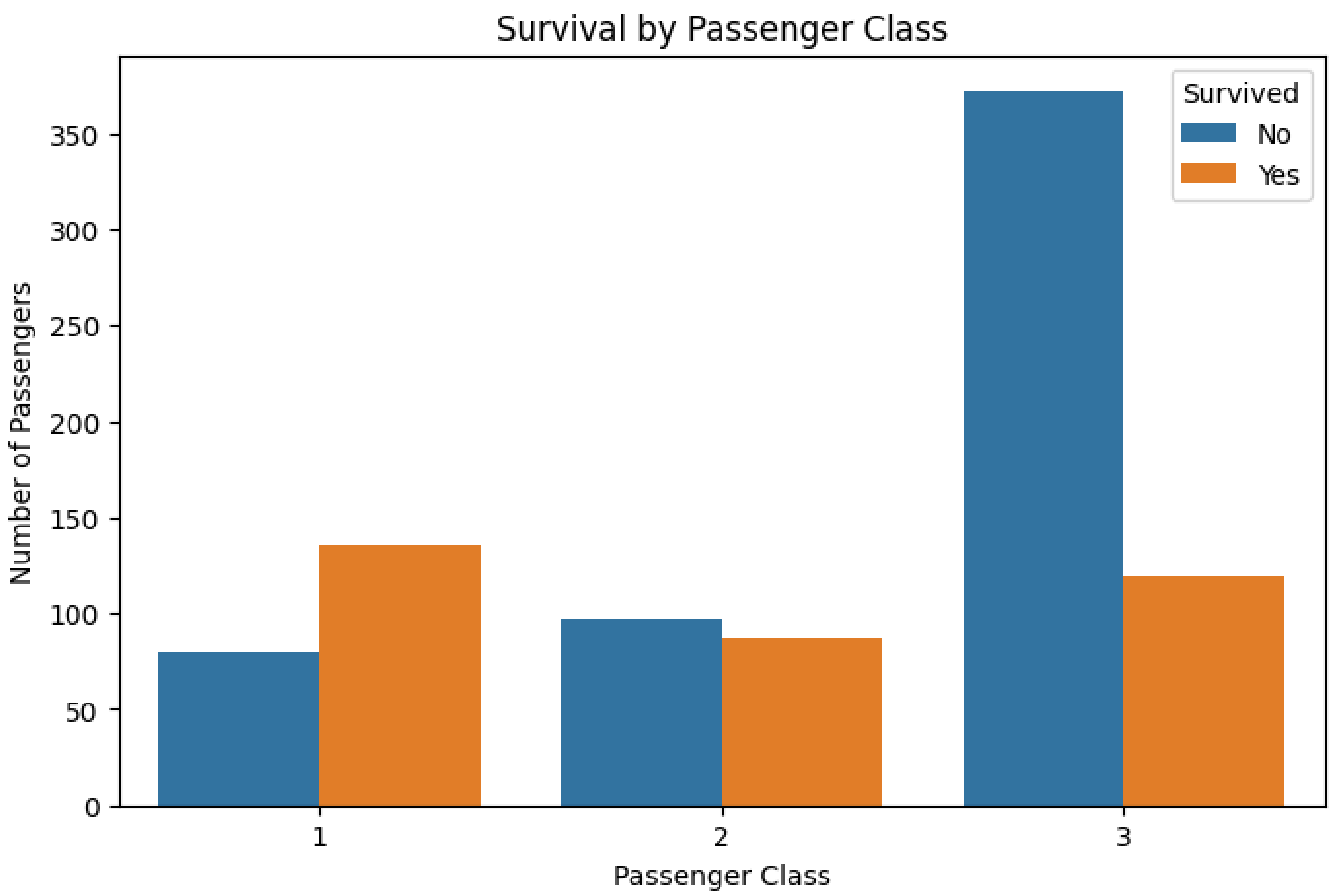


Observation: Female passengers had a much higher survival rate than male passengers. Most male passengers did not survive, while the majority of female passengers survived, showing a strong relationship between gender and survival.

4.2 Survival by Passenger Class

```
plt.figure(figsize=(8, 5))
sns.countplot(data=df, x='Pclass', hue='Survived')
```

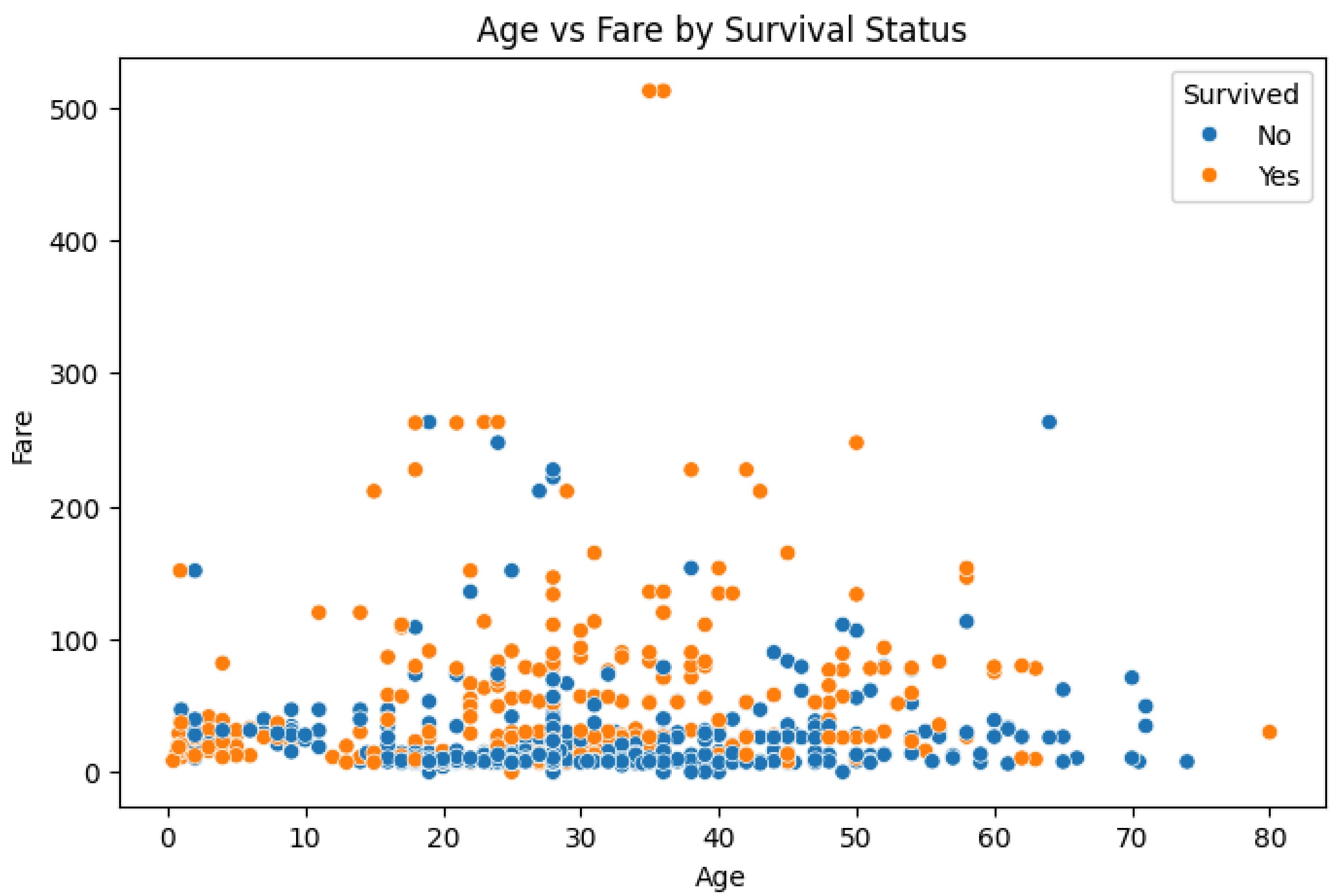
```
plt.title('Survival by Passenger Class')
plt.xlabel('Passenger Class')
plt.ylabel('Number of Passengers')
plt.legend(title='Survived', labels=['No', 'Yes'])

plt.show()
```



Observation: First-class passengers had a higher chance of survival, while third-class passengers experienced the highest number of deaths. This indicates a strong relationship between passenger class and survival.

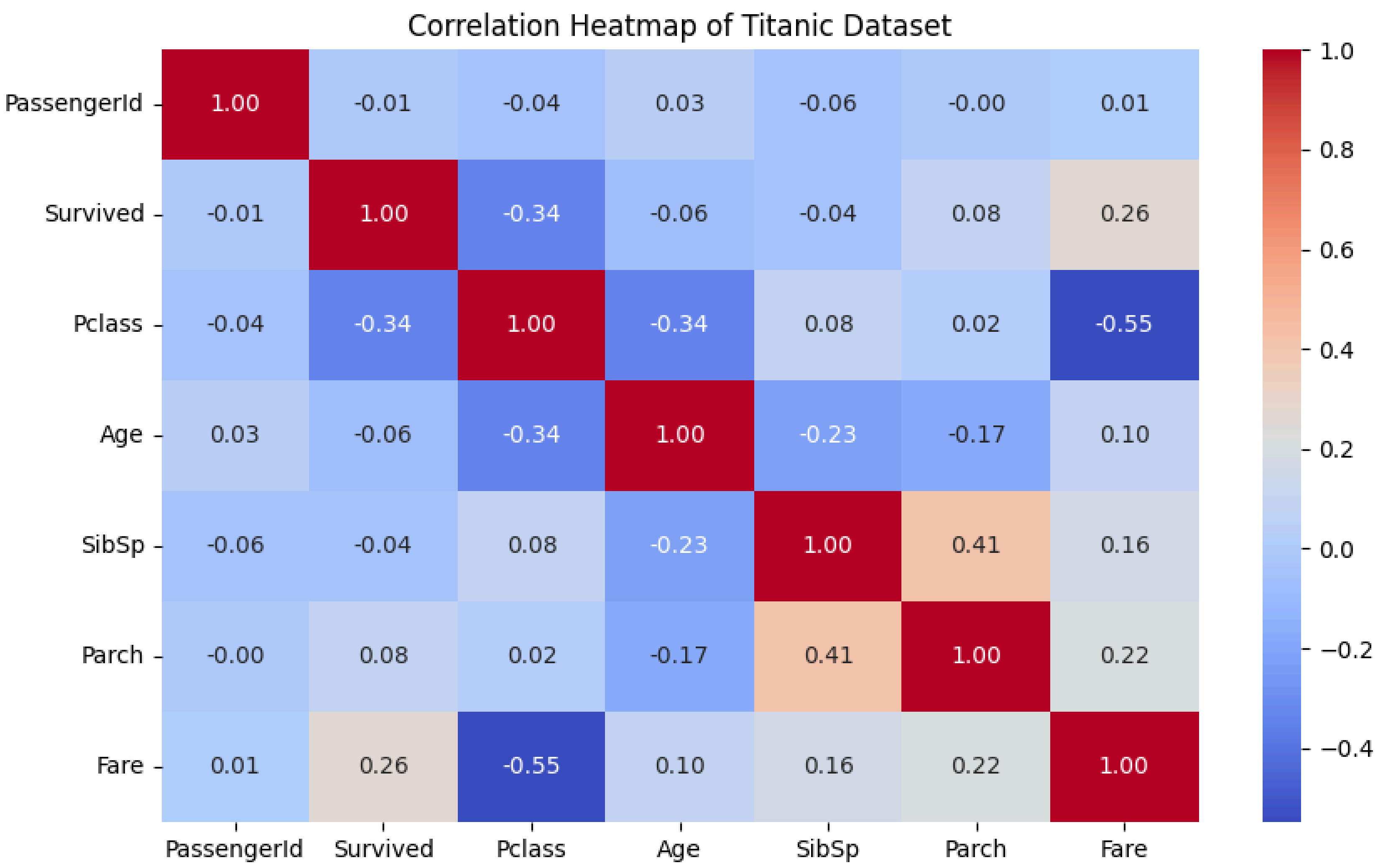
4.3 Age vs Fare

```
plt.figure(figsize=(8, 5))
sns.scatterplot(data=df, x='Age', y='Fare', hue='Survived')

plt.title('Age vs Fare by Survival Status')
plt.xlabel('Age')
plt.ylabel('Fare')
handles, labels = plt.gca().get_legend_handles_labels()
plt.legend(handles, ['No', 'Yes'], title='Survived')

plt.show()
```



Observation: There is no strong direct relationship between age and fare. However, passengers who paid higher fares appear more frequently among survivors, suggesting that fare and passenger class had an influence on survival.

```
numeric_df = df.select_dtypes(include=['number'])

plt.figure(figsize=(10, 6))
sns.heatmap(numeric_df.corr(), annot=True, cmap='coolwarm', fmt='.2f')
plt.title('Correlation Heatmap of Titanic Dataset')
plt.show()
```



Observation: Passenger class has a negative correlation with survival (-0.34), meaning first-class passengers were more likely to survive. Fare has a positive correlation with survival (0.26), while passenger class and fare have a strong negative correlation (-0.55). Age shows only a weak relationship with survival.

```
survival_by_gender = df.groupby('Sex')['Survived'].mean() * 100
```

```
print("Survival Rate by Gender (%):")
print(survival_by_gender)
```

```
Survival Rate by Gender (%):
Sex
female    74.203822
male      18.890815
Name: Survived, dtype: float64
```

```
survival_by_class = df.groupby('Pclass')['Survived'].mean() * 100

print("Survival Rate by Passenger Class (%):")
print(survival_by_class)
```

```
Survival Rate by Passenger Class (%):
Pclass
1    62.962963
2    47.282609
3    24.236253
Name: Survived, dtype: float64
```

```
overall_survival_rate = df['Survived'].mean() * 100

print(f"Overall Survival Rate: {overall_survival_rate:.2f}%")
```

```
Overall Survival Rate: 38.38%
```

5. Key Insights and Observations

- The overall survival rate was **38.38%**, meaning most passengers did not survive.
- Female passengers had a much higher survival rate (**74.20%**) than male passengers (**18.89%**).
- Passenger class strongly influenced survival. First-class passengers had the highest survival rate (**62.96%**), while third-class passengers had the lowest (**24.24%**).
- Most passengers were between **20 and 40 years old**